

8

Given 3×3 images X and Y below, each of which represented as an array of unsigned integers 8-bit (uint8). What would be the intensity value of the pixel at coordinate (2,3) in the output image $Z = X + Y$ using normalization technique ? 

$$X = \begin{bmatrix} 154 & 153 & 155 \\ 156 & 156 & 157 \\ 156 & 159 & 160 \end{bmatrix} \quad Y = \begin{bmatrix} 45 & 150 & 155 \\ 100 & 200 & 190 \\ 156 & 50 & 0 \end{bmatrix}$$

267

255

243

None of these

9

JPEG in image processing stands for 

Joint Photographic Experts Group

Joint Plate Expert Group

Joint Photographic Execute Group

None of these

10

If $n=200$ is a scalar that added to the image, then by subtracting n from the output image we will get the exact original image. 

☒ True

☐ False

11

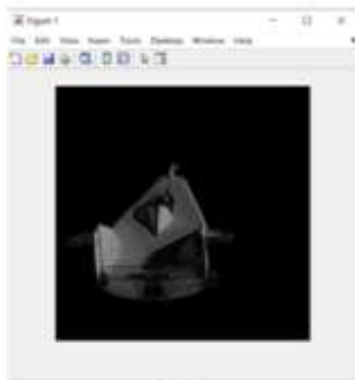
One of the common image processing operations is blurring which refers to the process of removing the blurring artifacts from images. 

☒ True

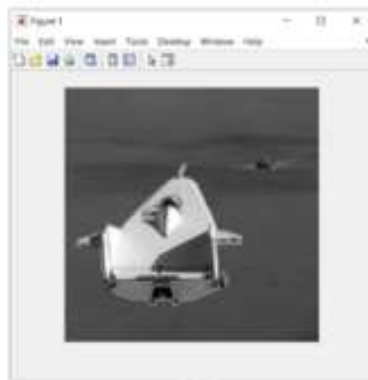
☐ False

Given an image I and its binary mask M below, each of which represented as an array of unsigned integers 8-bit (uint8). What would be the output of the following MATLAB code:

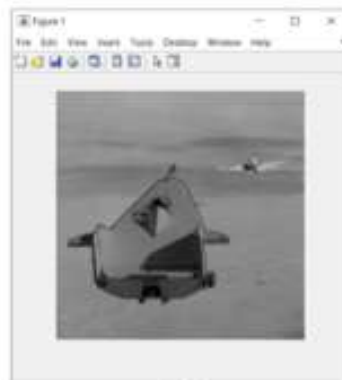
```
I2 = imdivide(I, 1.8);
M2 = bitcmp(M);
R1 = bitor(I, M);
R2 = bitor(I2, M2);
Output = bitand(R1, R2);
figure, imshow(Output)
```

 I  M 

(A)



(B)



(C)

None of these

(D)

Option A

Option B

Option C

Option D

13

Intensity levels in 8 bits image are 

☒ 128

☐ 255

☐ 256

☐ 512

14

Image manipulation is the process of making changes in images automatically based on carefully designed algorithms. 

☒ True

☐ False

15

Which operation is not pixel-by-pixel operation? 

Subtraction

Addition

Multiplication

None of these

16

Given a 3×3 image I , represented as an array of unsigned integers 8-bit (uint8). What would be the intensity value of the pixel at coordinate (2,2) after complementing this image? 

$$I = \begin{bmatrix} 161 & 159 & 153 \\ 156 & 154 & 157 \\ 158 & 160 & 155 \end{bmatrix}$$

17

A digital image is a representation of a two-dimensional image using a finite number of points. 

☒ True

☐ False

18

The first fundamental step in image processing is 

☒ Image segmentation

☐ Image acquisition

☐ Image enhancement

☐ Image restoration

What would be the output of the following MATLAB code:

```
Img_input = imread('cameraman.tif');  
image(Img_input)  
axis off  
axis image  
colormap('gray')
```



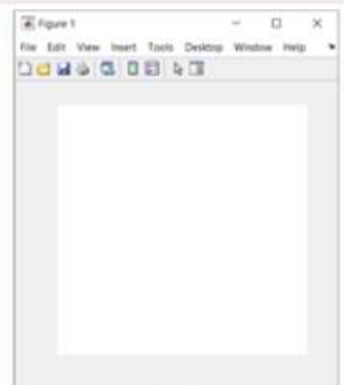
(A)



(B)



(C)



(D)

Option A

Option B

Option C

Option D

20

Logical operations between two or more images are performed on pixel-by-pixel basis, except for one that is performed on a single image. Which one is that? 

☒ AND

☐ OR

☐ NOT

☐ None of these

21

Each pixel in a digital image is represented by one or more numerical values. 

☒ True

☐ False

22

Given 3×3 images X and Y below, each of which represented as an array of unsigned integers 8-bit (uint8). What would be the intensity value of the pixel at coordinate (3,1) in the output image $Z = X \text{ AND } Y$?

$$X = \begin{bmatrix} 200 & 100 & 100 \\ 0 & 10 & 50 \\ 50 & 250 & 120 \end{bmatrix} \quad Y = \begin{bmatrix} 100 & 220 & 230 \\ 45 & 95 & 120 \\ 205 & 100 & 0 \end{bmatrix}$$

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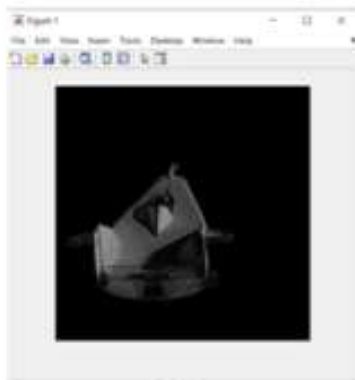
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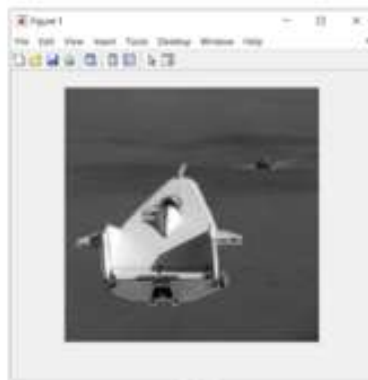
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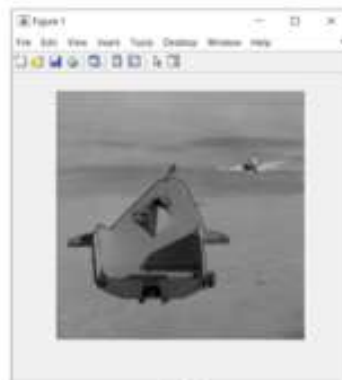
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Image segmentation

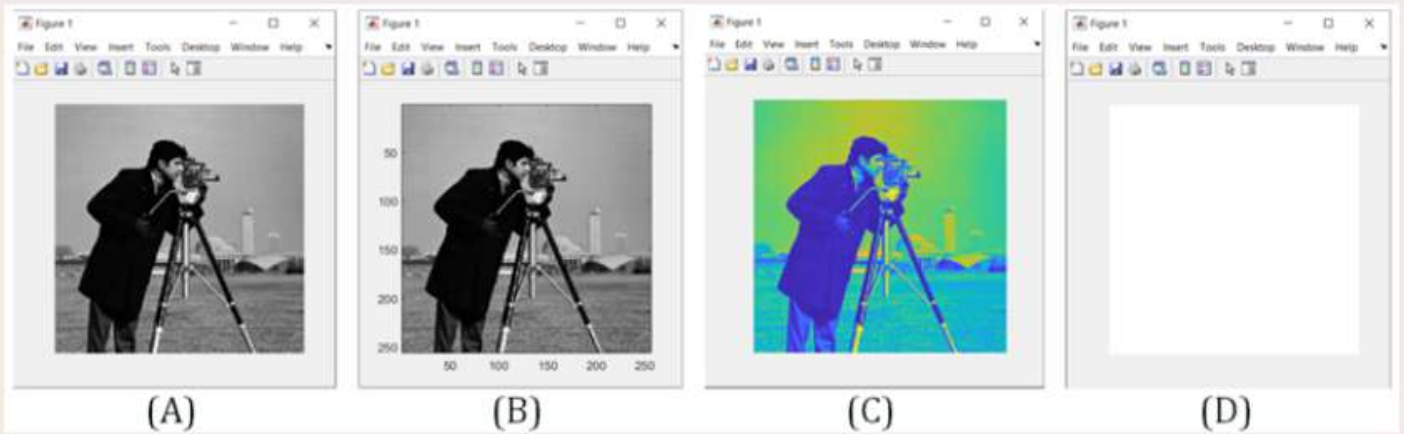
Image acquisition 

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Image restoration

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